



OHMGYM BY IRON CHAIN · CLOCK IN · THE SOLANA MOBILE HACKATHON · SUBMISSION OF OCTOBER 8, 2026

PROOF OF TRAINING, SIGNED BY THE FRIEND WHO WATCHED.

Iron Chain turns a workout into a proof on Solana. The user owns it. **No speculative token.**

Android

signed APK, MWA on a real device

1v1 live

a friend on the other phone

Apache-2.0

open proof protocol

Dahu Wake Family Sàrl · Geneva, Switzerland · ironchain.studiodahu.ch · [@ironchain_app](https://twitter.com/ironchain_app) · github.com/ironchain-app/ironchain-protocol



WHERE WE COME FROM

"WHAT MAKES A BOX ISN'T THE EQUIPMENT. IT'S THE PEOPLE."

We trained in a box in France, then moved away — and so did our friends. What we missed was never the equipment: the energy of people training beside you, a coach whose advice landed, a WOD scaled to the level you were actually at. No recorded programme can put two people in a session together.

So Joris built the app we could not find: **one that lets you train apart without training alone.** The session is scaled to you, on the gear you own, and a friend is on the other phone. That is why we wrote the live 1v1 ourselves — and went head to head with friends in other countries. **A leaderboard does not make you finish the set; the mate who pushes you and gives you grief does.**

Then came the part we had not planned for. That mate had watched you do it — and nobody had a way to prove it.

WHAT WE WANTED BACK

The energy of your friends, a coach whose advice lands, and a **WOD scaled to the level you are at** — in a garage, with one kettlebell.

WHAT WE BUILT, UNFUNDED

ohmGym. 240 commits since late August, 1,100 tests, live 1v1 video training.

WHAT THE WITNESS UNLOCKS

Iron Chain on Seeker. Your mate countersigns with a fingerprint in the Seed Vault — we know no other Solana app that does it. Set a goal, hit it, and the marker is yours for good: nobody can take it back, and nobody had to believe you.

THE PROBLEM

OTHER APPS PAID PEOPLE TO MOVE, TOOK THEIR WORD, **AND KEPT THEIR HISTORY**

THEY PAID PEOPLE TO MOVE

StepN and Genopets paid a token per step. When the money stopped, the people stopped. The ones still standing now live on insurers and corporate programmes, paid in ordinary currency.

Left with: **a dead token**

THEY TAKE YOUR WORD

Strava records what you declare, for 195 million registered athletes. A public link lets anyone look at a session. Nothing lets them verify it.

Left with: **a screenshot**

THEY KEEP YOUR HISTORY

Ten years of sessions belong to the company that stored them. Change app, and the record does not come with you. There is nothing to export that anyone else would accept.

Left with: **nothing at all**

The fix is not a better reward. It is a record the user owns, that someone else can check.

Networks built on physical work do not win by handing out tokens. They win when someone needs that work verified, and pays to read it.

WHAT IRON CHAIN IS

A TRAINING APP THAT WRITES **PROOF YOU OWN TO SOLANA**

You train alone, or head to head with a friend, one phone each.
When the clock stops, you both sign. The proof lands in your wallet, not in our database.

If we shut down tomorrow, you keep it. A stranger can verify it with a link.

The protocol underneath is open under Apache 2.0. Any app can issue and verify the same proofs.



Design target - the shipped screens are in the demo video

A human watching you train is the hardest witness to fool. The peer co-signs the proof.

SIX SIGNALS, NOT A SCORE

CODE	SIGNAL	WHAT IT ESTABLISHES	COST TO FAKE
D	Declared · the user logs the session	Nothing. It is the starting point.	None
B	Bracketed · two in-app photos, hashed on the device	Something happened between t_0 and t_1	Low to medium
M	Measured · heart rate, calories, steps over the session	A body was under load	Medium
C	Coherent · the heart-rate curve matches the declared effort	The effort matches the claim	High
W	Witnessed · a peer co-signs, present over live video	A human saw the session	Very high
N	Notarized · a verified coach co-signs	A professional stakes their reputation	Highest

"signals": ["D", "B", "M", "C", "W"] · A proof carries the signals it actually has. Nothing more. The reader applies its own threshold.

EVERYONE ELSE STOPS AT D

Every fitness-on-chain project we know writes a self-declared session and calls it proof. We record D as what it is: a claim. Then we add the rest.

WHY SIGNALS, NOT A SCORE

A "level 3" ages badly. The day one signal is faked, every past attestation collapses with it. Signals don't: stop trusting B, and M, C, W stay readable on proofs issued years ago.

W is the signal Iron Chain is built around. A watch can go on someone else's wrist. A person watching you cannot. We know no other Solana app that puts a human witness in front of a session.

FROM A SESSION TO SOMETHING **A STRANGER CAN CHECK**

1 · DEVICE

THE SESSION ENDS

The app assembles a leaf: window, activity, the signals actually present, and hashes. No health data, no photo, no name.

OFF-CHAIN

2 · WALLET

THE USER SIGNS IT

Mobile Wallet Adapter hands the leaf hash to Phantom, Solflare or the Seed Vault. The user approves on their own screen. We never hold the key.

OFF-CHAIN

3 · DEVICE

THE LEAF STAYS WITH THEM

Signed, stored on the phone and in their encrypted backup. It needs no network, so a basement gym is the normal case.

OFF-CHAIN

4 · SOLANA

ONE WRITE A MONTH

The Merkle root of the month is attested on the Solana Attestation Service, one attestation per person. A thousand users training fifteen times means a thousand writes, not fifteen thousand.

ON-CHAIN

5 · ANYONE

A STRANGER VERIFIES

Give a leaf, its inclusion path and the subject's key: recompute the root, compare. No account, no API, no permission, and no trust in us.

PUBLIC

The app never holds a key, and never holds the proof. Both belong to the person who trained, and both outlive us.

WHAT WE PUT ON SOLANA, **AND WHAT WE REFUSED TO**

THE SESSION NEVER GOES UP. THE MONTH DOES.

A signed leaf per session, kept on the device. One attestation per person per month carries the Merkle root of them all.

Why: **a thousand writes a year instead of fifteen thousand**, and a proof that still verifies one by one.

NO HEALTH DATA. NO PHOTO. DIGESTS ONLY.

Heart rate, calories and images stay on the phone. Only their SHA-256 reaches the chain, inside a hash that proves nothing about the body.

Why: **regulated data on a public ledger is disqualifying** on the dApp Store, and irreversible.

SIGNATURES TRAVEL IN THE DATA, NOT IN A PROGRAM.

The witness countersignature rides inside the attested payload and is checked off-chain by the open verifier.

Why: **no custom on-chain program to write** — nothing to exploit, nothing to audit, nothing to upgrade.

Every one of these costs us something, and we took it anyway. Monthly anchoring delays a proof. Hashing only means a reader must ask the user for the payload. No program means no on-chain enforcement. We preferred a design a stranger can verify over one we would have to be trusted for.

WHAT IT IS BUILT ON, **AND WHY THAT ONE**

FLUTTER & **DART**

One codebase for iOS and Android, with native bridges written in house for the wallet and the sensors. 240 commits, 1,100 automated tests.

MOBILE WALLET ADAPTER **2.0**

The Solana Mobile standard for signing on Android. Phantom and Solflare both connect on a physical device. Community-maintained Flutter client, not an official SDK — we say so because it matters for maintenance.

METAPLEX **CORE**

Permanent badges as soulbound assets. Measured on devnet against a tokenized attestation: less than half the rent, same two guarantees — the holder cannot transfer, the issuer can revoke.

WEBRTC

Peer-to-peer video for the live 1-on-1, with a relay only when two phones cannot reach each other. The duel is the product, so it could not be a third-party room.

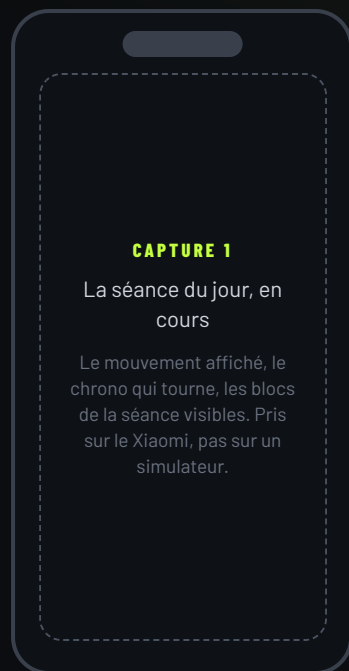
SOLANA ATTESTATION **SERVICE**

We define SAS schemas rather than invent our own registry. Maintained by the Solana Foundation, Apache-2.0, and it exists nowhere else.

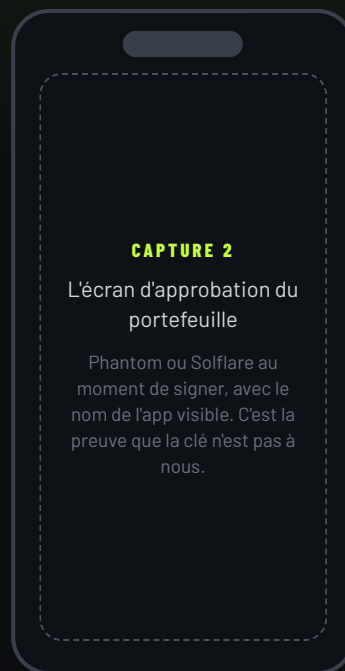
SEED VAULT **ON SEEKER**

Hardware-backed signing. A countersignature that costs a fingerprint exists only on Solana Mobile — that is why this app ships here before any app store.

THREE SCREENS, ONE PHYSICAL ANDROID



THE SESSION, RUNNING



THE WALLET ASKS, THE USER APPROVES



THE PROOF, AND HOW TO CHECK IT

Screenshots from the shipped Android build, not mockups. Every image on this page comes from the signed release APK running on a Xiaomi under Android 16 — the same build linked in this submission.

WHAT WAS BUILT WHEN

BEFORE THE HACKATHON, ON OUR OWN FUNDS. DURING CLOCK IN, THE ANDROID AND SOLANA EDITION.

COMPONENT	STATUS	PLATFORM
Workout engine, social feed, live 1v1 video over WebRTC, three duel modes, health sensors, nutrition	BEFORE - IOS, SELF FUNDED	ohmGym by Iron Chain - iOS, TestFlight
Android edition: signed release APK, full workout flow on a physical device	CLOCK IN - SEPT 17	Android
Mobile Wallet Adapter 2.0: Phantom and Solflare connect on a physical device	CLOCK IN - SEPT 17	Android
First session proof on-chain, as a signed memo through the MWA wallet	CLOCK IN - SEPT 14	Android - devnet
ironchain-protocol, the open standard, Apache 2.0	CLOCK IN - SEPT 15	public
SAS credential and ironchain-month schema deployed, addresses public, cost measured on a bench	CLOCK IN - SEPT 24	devnet
1v1 duel between two Android phones	CLOCK IN - WEEK OF SEPT 28	Android
In-app countersignature of a duel result by both players, verifiable	CLOCK IN - WEEK OF SEPT 28	Android
Seed Vault signing, the app writing attestations on that schema, photo hashing, Health Connect, dApp Store listing, quests	NEXT	Seeker

Repository created August 9, 2026. Rows marked "week of Sept 28" are updated at submission with what shipped. The first public release is the Solana dApp Store.

A WORKOUT IS THE REAL WORLD. IT TAKES A CAMERA, SENSORS, AND A FRIEND ON THE OTHER PHONE.

LIVE 1V1 VIDEO **W · WITNESSED**

Two friends train together over WebRTC. It works today. A watch can go on someone else's wrist. A person watching you train cannot be faked. The peer co-signs.

WEARABLE SENSORS **M · MEASURED** **C · COHERENT**

Health Connect reads calories, steps, resting heart rate and HRV. A claim becomes a measurement.

CAMERA CAPTURE IN THE APP **B · BRACKETED**

A photo at the start and at the end of the session, hashed on the device. Only the hash leaves the phone.

Together, these signals constrain each other. To fake a session you would need a heart rate curve between two photos taken in the app. You would have to train.



Design target · the shipped screens are in the demo video

WHERE EACH PIECE OF THE STACK STANDS TODAY

PIECE	STATUS	WHAT IT DOES FOR THE USER
Mobile Wallet Adapter 2.0	BUILT · SEPT 17	Phantom and Solflare connect on a physical Android device, on devnet. It is in the demo video.
Seed Vault	NEXT	Hardware signing on Seeker. A countersignature costs a fingerprint.
Who pays the fees	NEXT	The user pays from their Seeker wallet, or the premium plan covers it through a relay. The user sees a badge.
dApp Store	NEXT	Our launch platform. The signing key was generated on September 14.
Quests	NEXT	Our product is a badge system. Two quests at launch.

THE METRICS WE REPORT

dApp Store installs.

Attested sessions.

Duels co-signed by a live human witness, one phone each. We know no other Solana app that produces this number.

Instrumented from the first line of the integration.

ONLY ON SOLANA MOBILE

WHY SOLANA, AND WHY THE SEEKER

SOLANA ATTESTATION SERVICE

The attestation standard of this ecosystem, a public good under Apache 2.0. Our schemas extend it.

SEED VAULT

Hardware signing on the phone is a Solana Mobile primitive. A countersignature that costs a fingerprint exists only here.

OWNERSHIP THAT SURVIVES US

The attestation lives in the user's wallet. If Iron Chain shuts down, the proof remains and stays verifiable.

MOBILE WALLET ADAPTER

The Solana Mobile standard for connecting a wallet on Android. Phantom and Solflare connect on a physical device, built since September 17.

Together, these four mean **a duel needs no referee**. Two friends, two phones, two fingerprints — and a result neither of them can edit afterwards. The goal you set becomes a marker nobody can take back, **not even us**. None of that works if the record lives in our database.

We built ohmGym on iOS first. **The live 1v1 changed the plan**: a session witnessed in real time needs a chain that can carry the proof and a phone that can sign it. So **Solana Mobile gets this app first — before any app store**, and the ecosystem is where it can go further than we could alone.

Only a hash goes on-chain. Health data and photos stay on the phone. Swiss company, Swiss and EU data protection law.

THE CO-SIGNED DUEL, SIGNED IN THE SEED VAULT

During the hackathon, both players co-sign the duel result in the app. Next, that countersignature goes on-chain and into hardware.

Both players sign the result in the Seed Vault. A fingerprint each. One attestation carries both signatures, on the `duel-result` schema, on mainnet.

Health Connect adds calories, steps, resting heart rate and HRV. A claim becomes a measurement.

A duel starts from the lock screen, through a push notification. On screen, the countersignature reads “Certify Matt’s session”: one tap, one fingerprint.

A human witness and a hardware signature on each side. A reason for two people to own a Seeker.



Design target · the shipped screens are in the demo video

ACCEPTED WHEN

A duel between two Seekers produces one attestation on mainnet, co-signed by both through the Seed Vault, verifiable with the open verifier. Health Connect signals appear on devices that expose them.

Planned for the Seeker release, after the hackathon.

THREE PEOPLE, ONE SHIPPED PRODUCT, ONE ECOSYSTEM GRANT ALREADY DELIVERED

The hard problem is human. Our head of product comes from anthropology and social work.

JORIS BRUCHET

CTO and lead developer · Geneva · managing director · github.com/Superjojo74

Built the product: 240 commits since late August, 1,100 tests, live 1v1 WebRTC, a signed Android release, MWA on hardware. Native iOS bridges written in house.

20 years of production software. **Flutter apps live on the App Store and Google Play**, the latest in 2025.

Set up and ran an Ethereum validator node for his own company for three years. Solana and Ethereum.

Nine years of critical systems at the International AIDS Society, four World AIDS Conferences, 23,000 attendees. That is why we treat health data the way we do.

MÉLIBÉE BRUCHET

Head of strategy, product and operations · Geneva

Master's in Humanitarian Action, University of Geneva, and in Anthropology; state-qualified social educator. Five years in social work before product. Why people keep training, or stop, is her field.

Co-founded and ran Dahu Wake Park for seven years: a watersports and accommodation business. Real customers, partnerships, operations.

Full product lifecycle on web and mobile at Studio Dahu since 2022: framing, user journeys, delivery.

Runs the tester pilot, the funding applications, and the moderation and community-safety policy for the dApp Store launch.

LÉO DELION

Web3 product and economy · France · github.com/leo2lion

Won and delivered a Web3 ecosystem grant: TrustNomiks, backed by Intuition Protocol. 337 commits, deployed.

Cofounder of Nomiks, about €1M pre seed with Outlier Ventures. Economic modelling for Chiliz, The Sandbox, FDJ, Stacks.

Owns the open protocol: schemas, TypeScript SDK, integration docs. No speculative token.

Should the protocol ever need an ecosystem token, it would be decided after the product exists, issued under EU and Swiss rules, and never distributed as a reward for training.

WHAT EXISTS TODAY, AND WHAT STARTS NOW

The app, **ohmGym by Iron Chain**, runs on iOS TestFlight, internal. Iron Chain is the open standard underneath it. The first public release is the dApp Store.

The Android edition is signed with a dedicated dApp Store key and runs end to end on a Xiaomi, Android 16, since September 17.

MWA connects Phantom and Solflare on hardware. A finished session is written on devnet as a signed memo since September 14. Public protocol repository since September 15.

How it spreads. A duel takes two, so every player brings a partner. Every proof comes with a public link anyone can check: the Strava screenshot, made verifiable. Sharing your progression comes next.

The pilot is under way, self funded: ten testers who train in pairs for two weeks. We will publish the first results in early October, whether they are good or bad: sessions completed per tester, duels played, and who stopped and why.

HOW TO RUN IT

APK: the signed release in the submission. Any Android phone; a Solana wallet app in devnet mode (Phantom or Solflare).

Repository: `ironchain-app/ohmgym`, read access for the judges, build instructions in the README, no secret needed to compile.

Demo, 3 minutes, on a real device: a session, a duel, Finish, the signature in the wallet, the transaction on the explorer.

Live 1v1 with the team on request: the witness signal, experienced rather than read.

WHAT THE ECOSYSTEM KEEPS IF IRON CHAIN DISAPPEARS

We publish an open standard for verifiable physical activity under Apache 2.0: the proof-of-activity schema family on the Solana Attestation Service, plus TypeScript and Flutter SDKs. Public since September 15 at github.com/ironchain-app/ironchain-protocol.

The standard records **which independent signals corroborate a claim**: declared, bracketed, measured, coherent, witnessed, notarised. The reader applies its own threshold. A standard that hardcodes a level dies the day that level is defeated. One that records signals stays useful.

WHO CAN USE IT WITHOUT US

A fitness app whose users keep their history when they switch.

A challenge between friends that settles itself, with no referee.

A club or a community running a season nobody has to police.

A race organiser or a tournament that co-signs results.

Any Solana app that needs proof a human did something physical.

RUN IT, READ IT, CALL US.

CLOCK IN · the Solana Mobile hackathon · submission of October 8, 2026

THE APP — PLEASE TRY IT

Signed Android APK, direct download in the submission. Install it and run a full session.

On an iPhone? A TestFlight invite to ohmGym, our base app, on request. **The Solana layer is Android only.**

The duel needs two phones. Ask, and one of us takes you on, live.

Demo video, 3 minutes, on a real device.

THE CODE

`ironchain-app/ohmgym` — the app. Read access granted to the judges.

`ironchain-app/ironchain-protocol` — the proof protocol. **Apache-2.0, public since September 15.**

THE TEAM

Joris Bruchet · CTO, lead developer

Mélibée Bruchet · product and operations

Léo Delion · protocol and economy

Three people. No VC, no angel funding.

What we would do with a place on the dApp Store: launch there first, with two quests built on what the product already is — a badge system. Then bring the witness into the Seed Vault.



IronChain

Legal entity

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Links

ironchain.studiodahu.ch

github.com/ironchain-app/ironchain-protocol

[@ironchain_app](https://twitter.com/ironchain_app)

youtube.com/@ironchain-app

linkedin.com/company/studio-dahu

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